

S10 – Sensitivity Analysis – Specification Recoverability ROC Curve

Purpose

The purpose of this analysis is to visualise and quantify the ability of HIMS-CDI_cal to distinguish specification-derived DEPLOY_SYNTH profiles from NOT_DEPLOY_SYNTH profiles within the synthetic evaluation framework presented in Section 4 of the main manuscript.

Methodology

As described in Section 4.1 of the main manuscript, the calibrated multinomial logistic regression implementation (HIMS-CDI_cal) was evaluated using the stratified 48-sample synthetic test set consisting of:

- 16 DEPLOY_SYNTH samples
- 32 NOT_DEPLOY_SYNTH samples

Classification probabilities generated by the model were thresholded across the interval [0,1] to construct the Receiver Operating Characteristic (ROC) curve. The Area Under the ROC Curve (AUROC) was subsequently calculated using trapezoidal numerical integration.

Because the evaluation framework is specification-internal, the reference “ground truth” corresponds to the governance specification itself rather than to real-world network telemetry. No operational traffic datasets were used in this experiment.

Results

- **AUROC:** 0.75 (bootstrap 95% confidence interval: [0.72, 0.78])
- **Sensitivity at cost-optimised threshold**
(10:1 penalty weighting for false negatives): 0.94
- **Specificity at the same threshold:** 0.68
- **Expected Calibration Error (post-Platt scaling):** 0.03

These values correspond directly to those reported in Table 5 of the main manuscript and characterise boundary-recovery performance under synthetic Gaussian noise conditions ($\sigma = 0.10$).

The elevated sensitivity value reflects a governance-oriented preference for maximising the identification of deployable configurations, thereby reducing the risk of falsely excluding clinically acceptable deployments.

Interpretation

The ROC analysis demonstrates that HIMS-CDI_cal achieves stable discrimination between specification-derived deployable and non-deployable configurations despite the presence of controlled synthetic perturbation noise.

Furthermore:

- The AUROC value indicates moderate-to-strong specification recoverability.
- The low Expected Calibration Error confirms that post-calibration probability estimates remain well aligned with observed outcomes.

- The asymmetric sensitivity profile aligns with governance priorities that favour the identification of safe deployment.

Collectively, these findings support the argument that the proposed framework functions as a specification-recovery artefact rather than merely as a conventional predictive classifier.

Conclusion

The ROC analysis confirms that HIMS-CDI_cal reliably separates specification-derived deployable and non-deployable profiles within the synthetic evaluation framework. Consequently, the framework demonstrates measurable specification recoverability, thereby satisfying an essential requirement for design-science-oriented governance artefacts.